

Polyploid phasing as a p -fold decomposition of a balanced flow, and the homogeneous space of alleles

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Abstract

The k -mer spectrum of p homologous chromosomes sequenced together is the arc multiplicity vector $m = \sum_i \text{spec}(H_i)$ of one pooled de Bruijn multigraph, and phasing is the recovery of the summands. We show that the labelled phasings — ordered p -tuples of circular sequences whose spectra add to m — are counted by a sum over the ordered decompositions $m = c_1 + \cdots + c_p$ into connected balanced points of the box $[0, m]$ of the circulation lattice, each term the product of the per-sequence BEST counts $N_{\text{seq}}(c_i) = t(c_i) \prod (d^+ - 1)! / \prod c_i(a)!$. This is the lattice sum of the preceding note with the strand involution removed and one constraint replaced by a p -fold partition. The symmetric group $\mathfrak{S}_p$ permutes the parts of a decomposition; it does *not* act on the pooled graph, and the natural map from the disjoint union of the haplotype graphs to the pooled graph is a covering only when the haplotypes share no k -mer, so there is no monodromy in the sense of covering spaces. What replaces it is a homogeneous space: at a biallelic site carried by a of the p haplotypes the local choice is an a -subset of the labels, $\mathfrak{S}_p / (\mathfrak{S}_a \times \mathfrak{S}_{p-a})$, and for b sites in distinct $(k+1)$ -windows on a repeat-free reference the labelled count is $\prod_j \binom{p}{a_j}$ exactly, which $|\text{Hom}(\pi_1, \mathfrak{S}_p)| = (p!)^b$ does not reproduce. Sites within one $(k+1)$ -window merge into a single multi-allelic site whose alleles are the distinct local haplotypes. Unlabelled phasings are multisets, counted by $\prod_c \binom{N_{\text{seq}}(c) + r_c - 1}{r_c}$ over the multiset of spectra, equal to the labelled count divided by $p!$ exactly when the parts are pairwise distinct. Dropping the requirement that all haplotypes have the reference length exposes the copy-number ambiguity through repeats: the pooled spectrum of two copies of ϕ X174 is also the pooled spectrum of two circles of 2496 and 8276 base pairs, and of two tandem-dimer circles, whose weight in the sum is $1/2$. Every count is verified on ϕ X174 with synthetic substitutions at $p = 2$ and $p = 3$ by an enumeration of walks that does not use the lattice.

Keywords. de Bruijn graph, haplotype phasing, polyploid, BEST theorem, circulation lattice, flow decomposition, homogeneous space, copy-number variation.

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1 What was asked, and what is true

The preceding note [6] showed that a double-stranded reconstruction of a circular sequence S is a closed walk through half the arcs of the reverse-complement double cover whose image under

the strand involution ρ walks the other half, and that the count is a sum over the box points of the (-1) -eigenlattice of ρ on circulations. The involution was the obstacle: it reverses arcs, it is a duality rather than a symmetry of the digraph, and all the machinery of covering graphs is unavailable.

Polyploid phasing looks at first like the case where that obstacle is absent. The pooled k -mer spectrum of p homologous circular chromosomes $H_1, \dots, H_p$ is the arc multiplicity vector $m = \sum_i \text{spec}(H_i)$ of the pooled de Bruijn multigraph, and a phasing is a way of writing m as a sum of p spectra of circular sequences. Exchanging two haplotypes is a permutation of the summands; it does not reverse anything. One is tempted to say that the symmetric group $\mathfrak{S}_p$ acts on the pooled graph as a genuine automorphism group, that the haplotypes are the sheets of a p -fold cover, and that the phasing space is the space of flat $\mathfrak{S}_p$ -connections $H^1(G_k, \mathfrak{S}_p) = \text{Hom}(\pi_1(G_k), \mathfrak{S}_p)$ on the pooled graph, with a transposition at each heterozygous site. The specification of this note said so, and so did its planning.

None of that is so, and the correct statements are simpler. $\mathfrak{S}_p$ acts on the set of *decompositions* of m , not on the graph: the pooled graph has no distinguished symmetry exchanging haplotypes, because the haplotypes are different words and the graph does not remember which arc came from which. The natural map from the disjoint union of the haplotype graphs to the pooled graph is a covering map only when no k -mer is shared, in which case the pool is the disjoint union and there is nothing to phase (Proposition 7). And the local structure at a biallelic site is not a group but a homogeneous space: if a of the p haplotypes carry one allele and $p - a$ the other, the choice at the site is which a labels carry it, a point of $\mathfrak{S}_p / (\mathfrak{S}_a \times \mathfrak{S}_{p-a})$ of cardinality $\binom{p}{a}$; only for $p = 2$, or at a site where all p alleles differ, is this the group. For b sites in distinct $(k+1)$ -windows on a repeat-free reference the labelled count is exactly $\prod_j \binom{p}{a_j}$ (Theorem 8), and $(p!)^b$ is wrong already at $p = 3, b = 1, a = 1$: the answer is 3, not 6.

What survives, and is the subject of Section 2, is the lattice sum. The labelled phasings are counted exactly by

$$\#\text{Lab}(H_1 + \dots + H_p, k) = \sum_{\substack{c_1 + \dots + c_p = m \\ c_i \in Z_1 \cap [0, m], \text{ supp } c_i \text{ connected}}} \prod_{i=1}^p N_{\text{seq}}(c_i),$$

with N_{seq} the per-sequence BEST count of [6]; the unlabelled phasings by a multiset formula over the unordered decompositions (Section 4); and when the parts are not required to have the reference length the sum exposes the copy-number ambiguity through repeats, including tandem-dimer circles whose weight is $1/2$ (Section 5). Section 6 records the verification on ϕX174 .

Throughout, sequences are circular over $\{A, C, G, T\}$, a *circular sequence* is a word up to rotation, and $d(T)$ is the number of rotations fixing T . All computations are exact integer or rational arithmetic; the script `polyploid_phasing.py` reproduces every number below and prints a six-line pass/fail battery.

2 The pooled multigraph and the phasing lattice

Definition 1. Let $H_1, \dots, H_p$ be circular sequences of lengths $N_1, \dots, N_p$. The *pooled de Bruijn multigraph* $G_k = G_k(H_1 + \dots + H_p)$ has one vertex per k -mer occurring in some H_i and one arc per occurrence of a $(k+1)$ -mer in some H_i . A vertex is a *branch vertex* if it has at least two distinct successors or at least two distinct predecessors in the pool; the *compacted* pooled graph has the branch vertices as vertices and the maximal unitigs between them as arcs. An arc is

labelled by its haplotype and position; its *content* is the unitig string, A is the set of contents, $m(a)$ the number of arcs of content a , $\ell(a)$ its number of $(k+1)$ -mers, and $\partial: \mathbb{Z}^A \rightarrow \mathbb{Z}^V$ the boundary. $Z_1 = \ker \partial$ is the circulation lattice.

The compaction criterion differs from the one used for a single genome in [5], where a k -mer is a branch vertex if and only if it is repeated. In a pool every shared k -mer is repeated, so that criterion would contract nothing; the criterion of Definition 1 contracts exactly the k -mers through which every haplotype must pass in the same way. The per-sequence count below is intrinsic and does not depend on which compaction is used.

Definition 2. For $c \in \mathbb{Z}_{\geq 0}^A$ balanced with connected support let G_c be the multigraph with $c(a)$ labelled parallel arcs of content a , $t(c)$ its number of spanning arborescences converging on a root, $\text{Loc}(c) = \prod_v (d_c^+(v) - 1)!$ over the vertices of positive c -degree, and

$$N_{\text{seq}}(c) = \frac{t(c) \text{Loc}(c)}{\prod_a c(a)!} = \sum_{\text{spec } T=c} \frac{1}{d(T)} \in \mathbb{Q}_{>0},$$

the second equality being [6, Thm. 6]. $N_{\text{seq}}(c) = 0$ if the support is disconnected. The *length* of c is $|c| = \sum_a c(a)\ell(a)$.

Lemma 3. *Let T be a circular sequence all of whose $(k+1)$ -mers occur in the pool. Then T traverses each unitig of the compacted pooled graph a well-defined number of times, so $\text{spec}(T)$ is a vector $c \in \mathbb{Z}_{\geq 0}^A$ with $\partial c = 0$ and connected support, and $|c|$ is the length of T .*

Proof. An interior k -mer w of a unitig has exactly one successor and one predecessor in the pool. Every $(k+1)$ -mer of T through w is a $(k+1)$ -mer of the pool, hence is the unique in- or out-arc of w ; so each passage of T through w continues along the unitig in both directions, and the number of passages is the same at every k -mer of the unitig. $\square$

Definition 4. A *labelled phasing* of the pool at level k is an ordered p -tuple $(T_1, \dots, T_p)$ of circular sequences with $\sum_i \text{spec}(T_i) = m$ as multisets of $(k+1)$ -mers; its *weight* is $\prod_i 1/d(T_i)$. An *unlabelled phasing* is the multiset $\{T_1, \dots, T_p\}$. An *ordered decomposition* of m is a p -tuple $(c_1, \dots, c_p)$ of balanced vectors with connected support, $0 \leq c_i \leq m$ and $\sum c_i = m$; $\mathcal{D}_p(m)$ is their set. The *equal-length* phasings and decompositions are those with $|T_i| = N_1$, resp. $|c_i| = N_1$, for all i .

Theorem 5 (The phasing lattice). *The map $(T_1, \dots, T_p) \mapsto (\text{spec } T_1, \dots, \text{spec } T_p)$ is a surjection from the labelled phasings onto $\mathcal{D}_p(m)$, and the weighted number of labelled phasings is*

$$\#\text{Lab}(H_1 + \dots + H_p, k) = \sum_{(c_1, \dots, c_p) \in \mathcal{D}_p(m)} \prod_{i=1}^p N_{\text{seq}}(c_i).$$

The same holds for equal-length phasings with the sum restricted to equal-length decompositions. In particular the count is an integer whenever every part of every decomposition is the spectrum of aperiodic sequences only.

Proof. Lemma 3 sends each T_i to a balanced connected $c_i \geq 0$, and $\sum \text{spec } T_i = m$ is $\sum c_i = m$, which also gives $c_i \leq m$. Conversely each part of a decomposition is balanced and connected, so G_{c_i} has an Eulerian circuit, which read as a sequence has spectrum c_i ; the p sequences together have spectrum m . The fibre over $(c_1, \dots, c_p)$ is the product of the fibres over the c_i , each of weighted cardinality $N_{\text{seq}}(c_i)$ by Definition 2. Lengths are preserved by Lemma 3. $\square$

Remark 6. The lattice is the same object as in [6]: there $p = 2$, $c_2 = \rho c_1$ is forced, and the second part is the strand image of the first, so $\mathcal{D}_2(m)$ collapses to the box points of the anti-invariant lattice. Here nothing ties the parts together except $\sum c_i = m$, and $\mathfrak{S}_p$ acts on $\mathcal{D}_p(m)$ by permuting them. The single-strand theory is $p = 1$, where $\mathcal{D}_1(m) = \{m\}$ and the sum is one term.

Proposition 7 (No covering). *Let $\pi: \sqcup_i G_k(H_i) \rightarrow G_k(H_1 + \dots + H_p)$ send each vertex and arc of a haplotype graph to the same k -mer or occurrence in the pool. Then π is a surjective homomorphism of multidigraphs, and it is a covering map if and only if no k -mer occurs in two different haplotypes, in which case π is an isomorphism. In particular, when $p \geq 2$ and the haplotypes share a k -mer, $\mathfrak{S}_p$ acts on $\mathcal{D}_p(m)$ but there is no p -sheeted cover of the pooled graph with deck group $\mathfrak{S}_p$, and no action of $\mathfrak{S}_p$ on the pooled graph is induced by relabelling haplotypes.*

Proof. A covering map is a local isomorphism: the out-arcs of a vertex $\tilde{w}$ upstairs map bijectively onto the out-arcs of $\pi(\tilde{w})$. If a k -mer w occurs in H_i and in H_j , $i \neq j$, then w has two preimages $\tilde{w}_i, \tilde{w}_j$, the out-degree of $\tilde{w}_i$ is the multiplicity of w in H_i , and the out-degree of w is the sum of the multiplicities of w in all haplotypes, which strictly exceeds it. If no k -mer is shared, π is a bijection on vertices and on arcs. For the last sentence: an action of $\mathfrak{S}_p$ on the pooled graph induced by relabelling would have to fix every vertex and every arc, since the pool does not record labels, and the only such action is trivial. $\square$

The honest picture is therefore this. The pooled graph carries the flow m ; a phasing is a decomposition of the flow into p closed walks; the labels are a decoration of the decomposition and $\mathfrak{S}_p$ permutes the decoration. The question “how many phasings” is a question about the fibres of $\mathcal{D}_p(m) \rightarrow \mathcal{D}_p(m)/\mathfrak{S}_p$, and it is answered locally by homogeneous spaces (Section 3) and globally by a multiset count (Section 4).

3 The homogeneous fibre, and linkage

Fix a circular *reference* R of length N and let each H_i be obtained from R by substitutions at a set of positions, the *sites*; at a site s let the alleles be the letters occurring there across the H_i and $a_s(\alpha)$ the number of haplotypes carrying allele α . A site is *biallelic* if two letters occur, and we then write a_s for the count of the minor allele. Two sites are in the same $(k+1)$ -window if some $(k+1)$ -mer position of R contains both, i.e. their circular distance is at most k .

Theorem 8 (Independent sites on a repeat-free reference). *Suppose R has no repeated k -mer, the sites lie in pairwise distinct $(k+1)$ -windows, and no k -mer of any H_i overlapping a site coincides with a k -mer of R or with one overlapping another site. Then the equal-length ordered decompositions of m are in bijection with $\prod_s \mathfrak{S}_p / (\prod_\alpha \mathfrak{S}_{a_s(\alpha)})$, every part has $N_{\text{seq}} = 1$, and*

$$\#\text{Lab} = \prod_s \binom{p}{a_s(\alpha_1), a_s(\alpha_2), \dots} = \prod_{s \text{ biallelic}} \binom{p}{a_s} \quad \text{when all sites are biallelic.}$$

Proof. Under the hypotheses the compacted pooled graph is a cycle of N base pairs in which each site s is replaced by a *bubble*: the k -mer of R ending just before s is a branch vertex u_s , the k -mer starting just after s is a branch vertex v_s , and between them run parallel unitigs, one per allele α at s , each of length $k+1$ in $(k+1)$ -mers and of multiplicity $a_s(\alpha)$; the k -mers strictly between sites are shared by all haplotypes and contract into unitigs of multiplicity p . There are

no other branch vertices because R has no repeats and the allele k -mers are new. A closed walk in this graph has a winding number $w \geq 1$ around the cycle and length exactly wN , since all branches of a bubble have the same length. A part of an equal-length decomposition has length N , hence $w = 1$, hence passes each bubble exactly once through one branch; its spectrum is the 0/1 vector selecting one allele per site, G_c is a simple cycle, and $N_{\text{seq}}(c) = 1$. An ordered decomposition is thus a p -tuple of allele selections, one per label, with $a_s(\alpha)$ labels selecting α at each site: exactly the product over sites of the sets of ordered partitions of the labels into classes of sizes $a_s(\alpha)$, which are the homogeneous spaces stated, of cardinalities the multinomial coefficients. Theorem 5 gives the count. $\square$

Corollary 9. *Under the hypotheses of Theorem 8 the pooled graph has π_1 free of rank $\sum_s (n_s - 1)$, where n_s is the number of alleles at s , and $|\text{Hom}(\pi_1, \mathfrak{S}_p)| = (p!)^{\text{rank } \pi_1}$, which equals $\#\text{Lab}$ only when $p \leq 2$ or every site has p distinct alleles.*

Proof. The cycle rank of a cycle with bubbles is the number of extra branches. For $p = 2$ and biallelic sites both counts are 2^b ; for $p \geq 3$ and a biallelic site with $a_s = 1$ the site contributes p to $\#\text{Lab}$ and $p!$ to the other. $\square$

Theorem 10 (Linkage). *Let a set B of sites lie in one $(k+1)$ -window, with the hypotheses of Theorem 8 otherwise in force for B as a whole (no k -mer overlapping B is a k -mer of R or overlaps another site). Then in the compacted pooled graph B is a single bubble whose branches are the distinct local haplotypes — the distinct restrictions of $H_1, \dots, H_p$ to B — with multiplicity the number of haplotypes carrying each, and in Theorem 8 the block B counts as one site whose alleles are its local haplotypes. In particular two biallelic sites within distance k carried in only two combinations contribute $\binom{p}{a} \binom{p}{a'}$ and not $\binom{p}{a} \binom{p}{a'}$.*

Proof. Some $(k+1)$ -mer position of R contains all of B , so every haplotype has a $(k+1)$ -mer through the whole block, and two haplotypes with the same local restriction have the same k -mers throughout the window while two with different restrictions differ in that $(k+1)$ -mer. The k -mers overlapping B but not containing all of it may be shared between haplotypes that agree on the sites they contain; this merges vertices inside the bubble but creates no new closed walk of length N , because a walk of length N has winding number one and its passage through the bubble is determined by the $(k+1)$ -mer it uses at the position containing all of B . So the equal-length parts through the block are the local haplotypes, and the count is the multinomial in their multiplicities. $\square$

For sites that are pairwise within distance k but not all inside one $(k+1)$ -window, or a chain of sites each within k of the next, the bubble structure is a chain of partially merged bubbles and we do not have a closed form; see the open questions.

Remark 11 (With repeats). When R has repeats the argument of Theorem 8 gives a lower bound. Each allele selection $c \in \text{spec}(R)$ with one allele chosen at each site — is still a part of length N , and G_c is isomorphic to $G_{\text{spec } R}$ with the bubbles replaced by single unitigs of the same length, so $N_{\text{seq}}(c) = N_{\text{seq}}(R)$; the ordered allele selections therefore contribute $\prod_s \binom{p}{a_s} N_{\text{seq}}(R)^p$ to $\#\text{Lab}$. Other equal-length decompositions can exist only if $p \text{ spec}(R)$ admits a balanced connected sub-vector of length N that is not an allele selection, which requires a coincidence among the lengths of the sub-circles of R through its repeats. For ϕX174 at $k = 12$ no such coincidence occurs and the bound is the count at every tested configuration (Table 1).

4 The unlabelled multiset formula and the $\mathfrak{S}_p$ -action

Proposition 12. $\mathfrak{S}_p$ acts on $\mathcal{D}_p(m)$ by permuting the parts; the stabiliser of $(c_1, \dots, c_p)$ is $\prod_c \mathfrak{S}_{r_c}$, where r_c is the number of parts equal to c . If every sequence with spectrum a part of the decomposition is aperiodic, the number of unlabelled phasings whose spectra form the multiset $\{c^{r_c}\}$ is

$$\prod_c \binom{N_{\text{seq}}(c) + r_c - 1}{r_c},$$

and the total number of unlabelled phasings is the sum of this over the orbits $\mathcal{D}_p(m)/\mathfrak{S}_p$. It equals $\#\text{Lab}/p!$ exactly when every decomposition has pairwise distinct parts.

Proof. An unlabelled phasing with spectrum multiset $\{c^{r_c}\}$ is, for each c , a multiset of r_c sequences drawn from the $N_{\text{seq}}(c)$ sequences with spectrum c (integers, by aperiodicity); multisets of size r from a set of size n number $\binom{n+r-1}{r}$. The labelled phasings over an orbit number $|\text{orbit}| \cdot \prod_c N_{\text{seq}}(c)^{r_c} = (p! / \prod_c r_c!) \prod_c N_{\text{seq}}(c)^{r_c}$, and $\binom{n+r-1}{r} = n^r/r!$ only for $r \leq 1$. $\square$

The distinction is not academic. For three haplotypes S, S', S differing at one site, the three ordered decompositions all have the same multiset of spectra, and the unlabelled count is $N_{\text{seq}}(S) \binom{N_{\text{seq}}(S)+1}{2}$, which is 6 for ϕX174 at $k = 12$, against a labelled count of 24; $24/6 = 4 \neq 6$.

Remark 13 (Periodic parts). When a part c is $2c'$ for a circulation c' and $G_{c'}$ is Eulerian, the sequences with spectrum c include the tandem dimers UU of the sequences U with spectrum c' , which have $d(UU) = 2$, and more generally every sequence with spectrum c may be periodic, in which case $N_{\text{seq}}(c)$ is a half-integer or worse. The weighted count of Theorem 5 remains correct, the multiset formula does not apply to such parts, and the script reports them separately. In the data they arise only when the equal-length constraint is dropped, and they are the dimer circles of Section 5.

5 Copy-number variation across repeats

The equal-length constraint encodes the substitution-only model. Without it, the sum of Theorem 5 ranges over all decompositions of the pooled flow into closed walks, and for a reference with repeats it contains decompositions in which a block bounded by two copies of a repeat is carried once by one haplotype and three times by the other, or in which the two haplotypes are two circles of different sizes. These are the copy-number and excision–integration ambiguities of the pooled spectrum, and the lattice sum enumerates them.

For ϕX174 at $k = 12$ the single-strand graph is the interleaved two-copy repeat of [4], $a X_1 b Y_1 a X_2 b Y_2$: two branch 12-mers a, b and four unitigs, with $N_{\text{seq}}(S) = 2$. The sub-circles $a X_i b Y_j$ have lengths 2496, 2890, 2638 and 2748, in two complementary pairs summing to $N = 5386$. The pool $S + S$ has $m = 2$ on each unitig, and its ordered decompositions with connected parts number 17: one with two parts of length N , and sixteen of the shapes

$$(2496, 8276), (2890, 7882), (2638, 8134), (2748, 8024), (4992, 5780), (5276, 5496)$$

and their transposes. The first four are a sub-circle $a X_i b Y_j$ of length ℓ and the complementary circle of length $2N - \ell$, which carries the two unitigs of the sub-circle once and the other two twice: a sub-circle excised from one copy of the genome and its content integrated into the other.

configuration	$ V $	$ A $	rank Z_1	box	conn.	decomp.	#Lab	#Unl
1 site	4	7	4	26	25	2	8	4
2 sites	6	10	5	40	39	4	16	8
3 sites	8	13	6	56	55	8	32	16
5 sites	12	19	8	122	121	32	128	64
600, 606 linked + 2 sites	8	13	6	56	55	8	32	16
{600, 606} vs {600}	4	7	4	26	25	2	8	4

Table 1: Diploid pools, equal length. “decomp.” is the number of equal-length ordered decompositions; the prediction is 2^b for b linkage blocks, $\#Lab = 2^b N_{\text{seq}}(S)^2$ and $\#Unl = 2^{b-1} N_{\text{seq}}(S)^2$, met in every row. The walk enumeration reproduces N_{seq} on every spectrum it finds and the same $\#Lab$; the explicit unordered pairs number 4, 8, 16, 64, 16, 4. Checks (P), (L), (W), (U).

configuration	$ V $	$ A $	rank Z_1	box	conn.	decomp.	#Lab	#Unl
three sites: $a = 1$, $a = 2$, triallelic	8	14	7	248	247	54	432	72
one site, haplotypes S, S', S	4	7	4	68	67	3	24	6

Table 2: Triploid pools, equal length. First row: sites 600 (minor allele in one haplotype), 1700 (in two) and 2900 (three alleles); $\binom{3}{1}\binom{3}{2}3! = 54$ decompositions, all with pairwise distinct parts, $\#Lab = 54 \cdot 2^3$ and $\#Unl = \#Lab/6$; $|\text{Hom}(\pi_1, \mathfrak{S}_3)|$ would be $6^3 = 216$. Second row: three decompositions with the same multiset of spectra, $\#Unl = N_{\text{seq}}(S) \binom{N_{\text{seq}}(S)+1}{2} = 6 \neq 24/6$. Check (T).

The last two have *both* parts periodic: $4992 = 2 \cdot 2496$ and $5780 = 2 \cdot 2890$, the tandem dimers of a complementary pair of sub-circles, each with $d(T) = 2$ and weight $1/2$. With one substitution added the decompositions number 24, of which 22 have unequal lengths (Table 3). A pooled spectrum thus does not distinguish two identical circular chromosomes from two dimer circles of complementary halves, and the count records the fact with the weight it deserves.

6 Verification and data

All configurations use ϕX174 (NC_001422.1, 5386 bp) as the reference at $k = 12$, where $N_{\text{seq}}(S) = 2$, with substitutions at positions 600, 1700, 2900, 3900, 4900 (pairwise distance $> k$, none within k of a repeat k -mer) and, for linkage, at 606. Each row of Tables 1–3 is computed twice: from the lattice, by enumerating the box $Z_1 \cap [0, m]$ with the balance constraint and evaluating N_{seq} at every point; and by an enumeration of closed walks T_1 in the content multigraph (rooted at the smallest content a_0 , weighted $1/u(a_0)$, usage bounded by m) that never sees the lattice, followed for $p = 2$ by an explicit enumeration of the partner walks T_2 and a count of the unordered pairs $\{T_1, T_2\}$ as sets.

The six checks. (P) the pooled compacted graph is balanced and gains exactly two branch vertices per site away from repeats; (L) b unlinked biallelic sites give 2^b equal-length decompositions and $2^b N_{\text{seq}}(S)^2$ labelled phasings, and two sites within k are one site; (W) for $p = 2$ the walk enumeration reproduces N_{seq} on every spectrum and $\sum_{T_1} N_{\text{seq}}(m - \text{spec } T_1)$ equals the lattice sum; (U) the unordered phasings enumerated as sets of explicit walk pairs equal the multiset formula; (T) for $p = 3$ the decompositions number $\prod_s \binom{3}{a_s}, 3!$ at the triallelic site, and

configuration	box	conn.	decomp.	unequal	periodic	#Lab
diploid, 1 site, all lengths	26	25	24	22	4	66
homozygous $S + S$, all lengths	19	18	17	16	4	33

Table 3: Without the equal-length constraint. “unequal” is the number of ordered decompositions whose parts have different lengths; “periodic” the number with a periodic part, each counted with weight $1/2$ per such part. Length profiles for $S + S$: one (N, N) , and $(2496, 8276)$, $(2890, 7882)$, $(2638, 8134)$, $(2748, 8024)$, $(4992, 5780)$, $(5276, 5496)$ twice each. The explicit pair enumeration for the diploid row finds 34 unordered pairs: 32 on the ten decompositions without a periodic part, matching the multiset formula, and two pairs of dimer circles. Check (C).

the unlabelled count is the multiset count; (C) free lengths admit copy-number decompositions through the repeat, and the equal-length unlabelled count of $S + S$ is $\binom{N_{\text{seq}}(S)+1}{2} = 3$. All six pass, in under a second.

7 Scope, provenance, and what is open

Scope. Theorem 5, Proposition 7 and Proposition 12 hold for arbitrary circular haplotypes. Theorem 8, Corollary 9 and Theorem 10 assume a repeat-free reference and isolated sites; Remark 11 is a lower bound for references with repeats, met with equality in all data. The model is the pooled k -mer spectrum alone: no reads, no read pairs, no coverage, no errors. Reads that span sites constrain the decompositions further and are the whole content of the haplotype assembly problem [9, 3, 2]; what is counted here is the ambiguity that remains before any read is used, which is the analogue for phasing of the reconstruction count of [8] for assembly.

Provenance. The BEST theorem is [1, 10], its per-sequence normalisation and the notation N_{seq} are [6], and the single-genome decomposition of ϕX174 at $k = 12$ is [4]. Haplotype assembly as a combinatorial problem on sites is [9]; the cycle-basis formulation of [2] is the closest in spirit to the circulation lattice here, though it lives on a graph of sites and reads rather than on the de Bruijn graph. Polyploid phasing methods are surveyed in [7]. We are not aware of prior work counting phasings through the de Bruijn graph, or of the homogeneous-space description of the local fibre; a targeted search found none.

Open. *First*, the count for a chain of sites each within k of the next but not all in one window (partially merged bubbles). *Second*, the equality in Remark 11: characterise the references for which $p \text{ spec}(R)$ has no balanced connected sub-vector of length N beyond the allele selections. *Third*, the copy-number decompositions at genome scale, where the lattice has the rank of Z_1 of the pooled skeleton and the sum is not enumerable; a transfer-matrix evaluation along the genome is the natural tool, as it is for the double-stranded sum of [6]. *Fourth*, the effect of reads: a read spanning sites s, s' is a linear constraint on the decomposition, and the phasings compatible with a read set are a sub-lattice section; whether the standard phasing objectives (minimum error correction, maximum fragments cut) have an expression in these terms is unexplored.

References

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